



Credit Risk Segmentation & NPA Reduction Strategy

GrayQuest Education Finance Pvt. Ltd.

About The Company

GrayQuest Education Finance

Understanding the business before the analysis

GrayQuest is a Mumbai-based fintech servicer that enables students to pay school and college fees in easy monthly instalments. It was founded by Mr. Rishabh Mehta in the year 2017. The company operates as a fee-financing platform and partners with NBFCs that provide the loan capital, while GrayQuest manages customer onboarding, servicing, and collections. With a presence across thousands of educational institutions, GrayQuest aims to improve access to education by making fee payments more flexible and affordable.

BUSINESS AT A GLANCE

Business Model

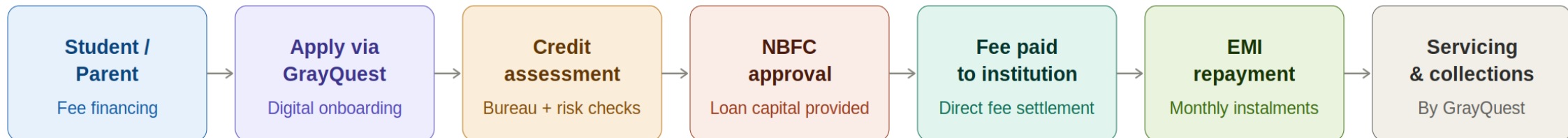
- Fee-based servicer
- Ties up with schools and colleges for solving the problem of fee collection
- Partners with NBFC lenders like Mirae Asset, Fintree, Akra Fincap, Western Capital, Ratnaafin Standard, etc
- FY25 Revenue: ₹24.6 Cr (+37% YoY)

Product

- 0-cost or low interest EMI : 6, 9 or 11 instalments
- School and college fees
- ~6,500 institutions nationally
- ~2 million borrowers



HOW GRAYQUEST OPERATES



Business Problem Statement

Why This Project Existed

The headline looked fine. The details told a different story. While the overall NPA rate appeared healthy, a deeper analysis uncovered concentrated pockets of risk driven by manual approval practices, repeat defaulter approvals, and NBFC-level performance differences. The project was undertaken to identify these hidden risk drivers and develop actionable strategies to improve portfolio quality and reduce future NPAs.

The Surface

1.34%

Portfolio NPA Rate — appears healthy

The Reality

1. Process Failure

The manual approval channel is 42.5% of loans but drives 74.7% of all defaults. The human override process is systematically undoing the credit algorithm.

2. No Safeguard

Prior GrayQuest defaulters (GQ_Defaulters) are being re-approved. They default at 7.36% — 5.5× the portfolio average.

3. NBFC Risk Concentration

Ratnaafin Lite NPA = 5.15% — 10× worse than Mirae Asset (0.51%). No risk-based NBFC routing exists.

Business Impact of Solving This: ₹55.4L avoidable NPA/cycle · 57 preventable defaults · **NPA from 1.34% → 0.91%**

Project Snapshot

Project Snapshot



Key facts before we begin

24,626	13,011	1.34%
Total Applications in Dataset	Disbursed Loans Analysed	Portfolio NPA Rate (Starting Point)
174	6	181
Total Unique Defaults	NBFC Partners Covered	Accounts in NPA File (₹1.03 Cr Outstanding)

PROJECT CONTEXT

The Dataset

- 2 institutions:
- Sri Chaitanya — 8,094 loans
- Narayana — 4,917 loans
- November 2025 NACH collection cycle included.

Dataset

FY 2024–25 data. Two institutions: Sri Chaitanya (8,094 loans) and Narayana (4,917 loans).

Deliverable

A credit risk scorecard, predictive model, early warning system, and actionable recommendations to reduce NPA from 1.34% to below 1%.

TOOLS USED

Python (statsmodels, scipy, pandas) · Microsoft Excel (pivot analysis, hypothesis testing, EWS framework) · Power BI (dashboard)

Project Objectives & Scope

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Clearly defined boundaries and measurable targets

PRIMARY OBJECTIVE

Build a credit risk scorecard to segment borrowers by default probability and reduce GrayQuest's NPA rate from 1.34% to below 1% within 6 months
— using only data already available at loan onboarding.

SECONDARY OBJECTIVES

Segment & Diagnose

Identify which borrower segments, approval channels and NBFC partners drive disproportionate NPA risk

Early Warning System

Design a post-disbursement NACH-trigger framework to intervene before defaults crystallise at 90+ DPD

Actionable Recommendations

Deliver specific, quantified, immediately implementable actions across underwriting, collections, NBFC routing and data systems

SCOPE

Covered

- 13,011 disbursed loans (FY 2024–25)
- Sri Chaitanya + Narayana only
- 6 NBFC partners | November 2025 NACH data

Excluded

- Pre-credit marketing funnel
- NBFC internal operations
- Revenue / P&L modelling

Structure of the Analysis

01

Data Preparation

Cleaned 24,626 application records. Identified 4,541 duplicates (18.4%). Isolated 13,011 disbursed loans as the analysis universe. Built NPA flag, customer profile labels and DPD buckets.

02

Descriptive Segmentation

Broke the portfolio by CIBIL band, income type, tenure, approval channel, NBFC partner and customer profile. Calculated NPA rates for each segment to identify where defaults concentrate.

03

Hypothesis Testing

Statistically confirmed which variables are genuine predictors ($p < 0.001$) and which are noise (loan amount, $p = 0.774$). Used Welch's t-test, chi-square, Z-test, ANOVA. Proved manual vs bureau gap exists even within the same CIBIL band.

04

Logistic Regression Model

Built a binary default prediction model. Validated with ROC-AUC = 0.787 (industry threshold ≥ 0.75) and McFadden $R^2 = 10.67\%$. Custom classification threshold of 0.055 set to handle the 1.34% event rate.

05

Scorecard + EWS + Recommendations

Translated regression output into a 4-variable scorecard (0–9 points), four risk tiers with decision rules, and an 8-trigger Early Warning System. Simulated 6 policy scenarios to identify optimal NPA reduction path.

Key Findings

The Manual Approval Channel Is the Primary Driver of Portfolio Risk

GrayQuest uses two approval routes: an automated bureau algorithm and a manual human override. The data reveals a stark and statistically confirmed divergence.

Manual vs. Bureau Approval Risk

	 Bureau Approval	 Manual Approval
Portfolio Share	58%	42%
Default Contribution	25%	75%
NPA Rate	0.59%	2.36%
Controlled Test (CIBIL 700-750)	1.39%	3.14%

2.36% vs **0.59%**
Manual NPA Rate Bureau NPA Rate

Manual approvals default about 4 times more often than bureau approvals.

3.14% vs 1.39%

The difference remains constant when we consider the most usual CIBIL band of 700–750, same borrower quality, different process resulting to the difference in NPA rate

Implication:

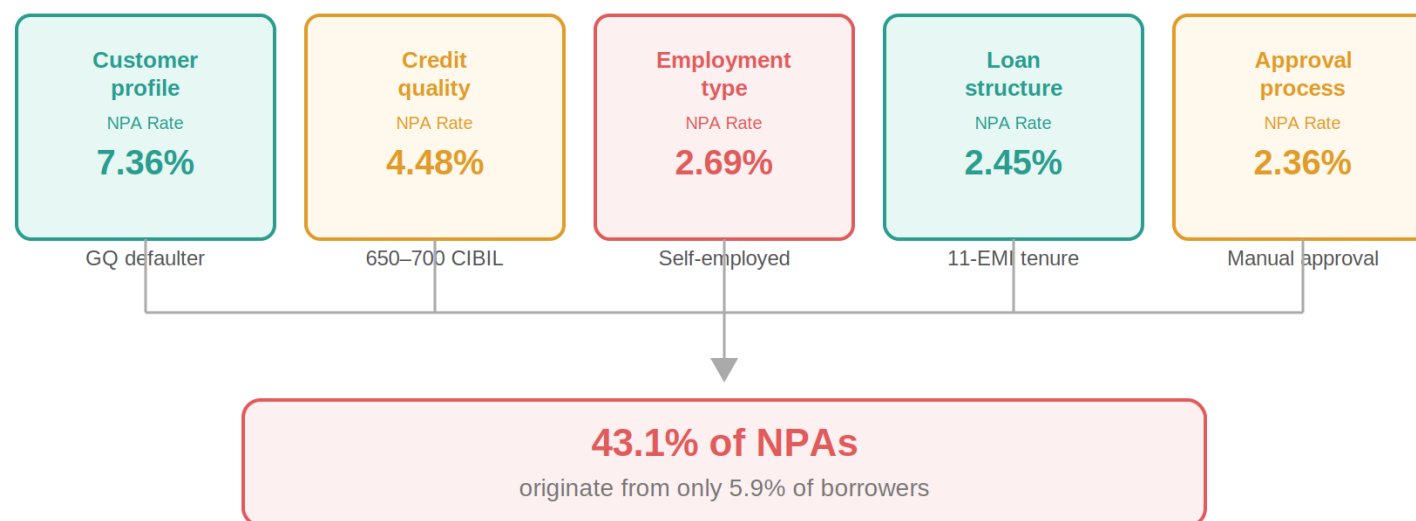
The 3.14% vs 1.39% within-CIBIL comparison confirms this is a process problem, not a borrower quality problem. Restricting manual overrides would prevent an estimated 97 excess defaults per cycle and eliminate ₹55.4L in avoidable NPA.

Default Risk Is Concentrated in a Small Number of Identifiable Borrower Segments

NPA is concentrated — not random — and identifiable at onboarding

Hypothesis testing (chi-square, ANOVA, Welch's t-test) confirmed five variables as statistically significant predictors of default ($p < 0.001$). Loan amount was not significant ($p = 0.774$) and was excluded.

Root causes of default



6-EMI + Bureau approval

0.43% NPA — safest possible combination

11-EMI + Manual approval

3.63% NPA — 8× higher than the safest cell

GQ Defaulter profile

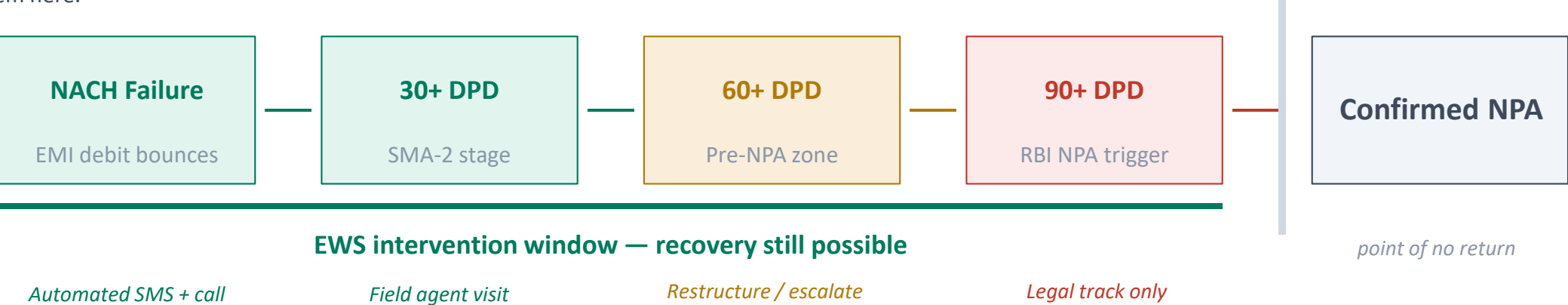
7.36% — should be auto-rejected; currently re-approved

While the Very High Risk segment accounted for only 5.9% of borrowers, it generated 43.1% of portfolio NPAs. This suggested that a targeted underwriting strategy could potentially deliver significant risk reduction without materially affecting overall business volume. To test this hypothesis, six approval scenarios were simulated using the scorecard to evaluate their impact on NPA reduction, portfolio quality, and business growth.

Default Follows a Predictable Journey, Creating Opportunities for Early Intervention

A predictable progression — and a recoverable portfolio

Every confirmed NPA passed through identifiable, trackable stages. 55.2% of at-risk accounts are still in the 30–60 DPD window — SMA-2 stage, not yet NPA. The EWS exists to catch them here.



DPD DISTRIBUTION — 181 ACCOUNTS | ₹1.03 CR OUTSTANDING

100	30–60 DPD (55.2%)	₹59.95L — still recoverable ✓
45	60–90 DPD (24.9%)	₹25.70L — act quickly
36	90+ DPD (19.9%)	₹17.54L — confirmed NPA

Default is not a sudden event. It follows a predictable progression — from NACH failure through delinquency to NPA — creating clear intervention opportunities.

Analysis

Credit Risk Scorecard



4 variables · 9-point scale · 4 decision tiers

Points are assigned in proportion to each variable's observed default rate. The scorecard is built entirely from data already captured at loan onboarding — zero new infrastructure needed.

POINT ASSIGNMENT

Variable	Category	Pts	NPA Rate
CIBIL	>750	0	0.73%
	700–750	1	2.35%
	650–700/NTC	2	4.48%
Income	Salaried	0	0.95%
	Self-Employed	1	2.69%
Tenure	6-EMI	0	0.43%
	11-EMI	2	2.45%
Channel	Bureau	0	0.59%
	Manual	3	2.36%

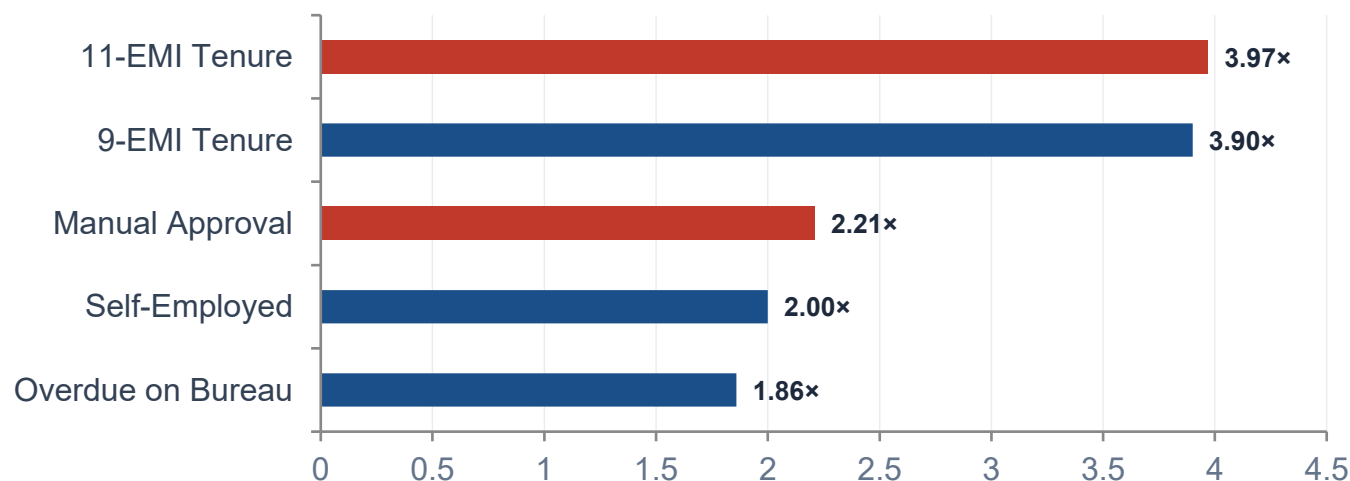
RISK TIER → DECISION RULE

Low Risk	Score 0–1	Auto-Approve
72.4% · 0.66% NPA		
Medium Risk	Score 2–3	Conditional Approval
18.1% · 2.08% NPA		
High Risk	Score 4–5	Enhanced Review
9.4% · 5.13% NPA		
Very High Risk	Score 6–9	Auto-Reject
5.9% · 6.04% NPA		

Logistic Regression — Predicting Default Probability

A logistic regression model was built to assign each borrower a probability of default. It was chosen over more complex models for interpretability — every rejection must be explainable to the borrower and defensible to regulators.

VARIABLE IMPORTANCE — ODDS RATIOS



Red = highest impact variables · Odds Ratio > 1 = increases default probability

ROC-AUC

0.787

The model successfully differentiates high-risk borrowers from low-risk borrowers with good accuracy.

McFadden R²

10.67%

The model captures key patterns behind borrower defaults and provides meaningful predictive insights.

Threshold

0.055

A customised risk threshold was used to ensure that potential defaulters are identified early rather than being missed by a standard rule.

Why threshold = 0.055, not 0.50?

With only 1.34% of borrowers defaulting, the default threshold of 0.50 would classify everyone as non-default and catch zero actual defaulters. The 0.055 threshold was calibrated to maximise the F1-score — balancing the cost of missing true defaults against flagging too many good borrowers.

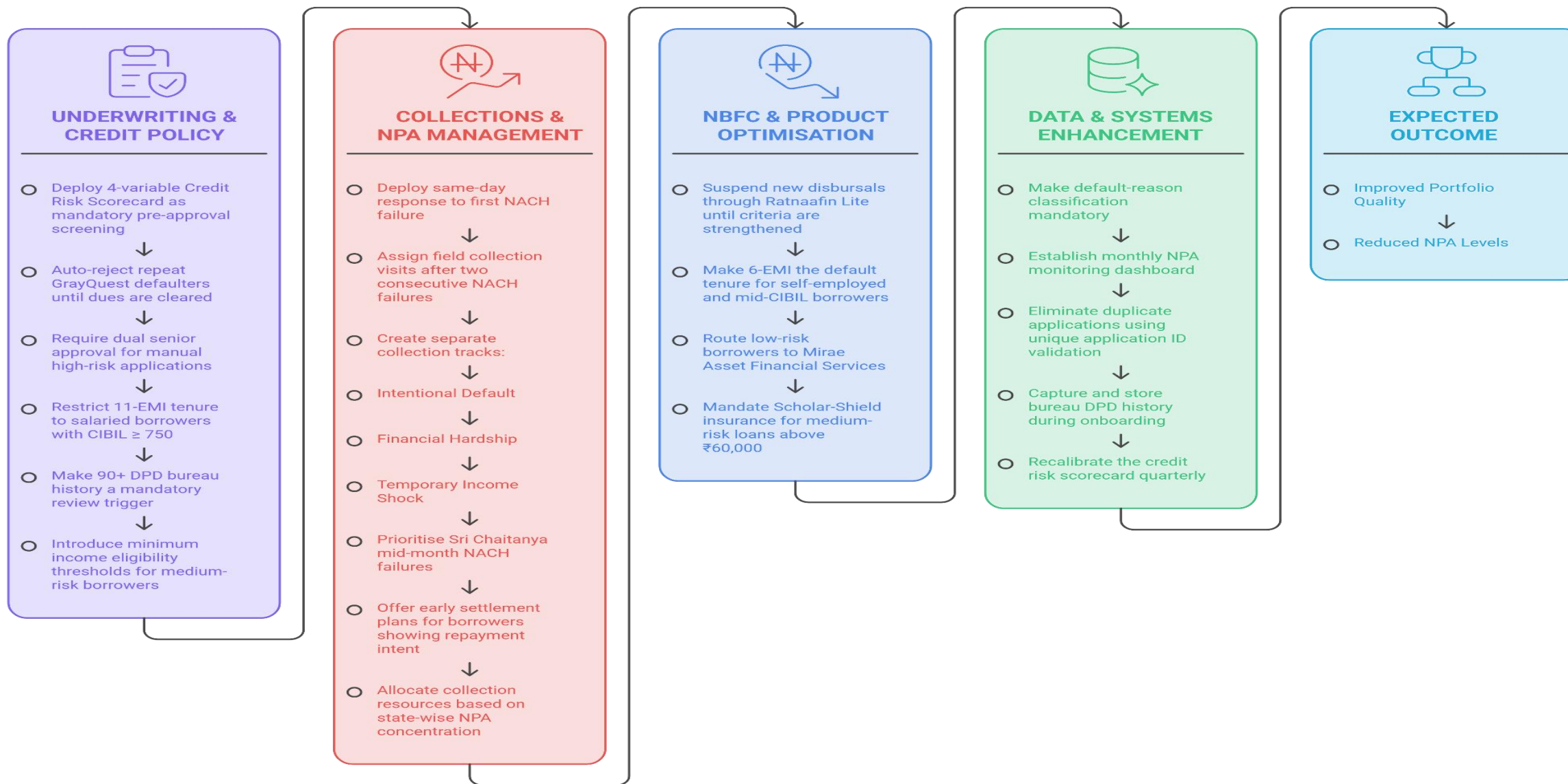
Tiered Approval Simulation — 6 Scenarios Evaluated

Each scenario balances NPA reduction against revenue foregone. Scenario F selected as optimal — as it gives the best NPA cut with manageable trade-off.

Scenario	Policy Rule	Rejected	Defaults Prevented	NPA Rate After	NPA Saved	Revenue Foregone	Signal
A Targeted	Reject GQ_Defaulters only	258 (2.0%)	19 (10.9%)	1.19%	₹10.8L	₹32.6L	Short-term win; strong symbolic impact
B Moderate	Reject CIBIL_Defaulters (manual) only	992 (7.6%)	44 (25.3%)	1.00%	₹25.1L	₹115.1L	Large revenue cost; meaningful NPA cut
C Combined	Reject GQ_Defaulters + CIBIL_Defaulters (manual)	1,250 (9.6%)	63 (36.2%)	0.85%	₹35.9L	₹147.7L	Near break-even with provisioning savings
D Score-Based	Reject all Score ≥ 5	992 (7.6%)	53 (30.5%)	0.93%	₹30.2L	₹118.4L	+ve long-term with ROE & provisioning adj.
E Full Tiered	Reject High + Very High (Score ≥ 4)	1,606 (12.3%)	73 (42.0%)	0.77%	₹41.6L	₹193.1L	Near break-even with provisioning savings
F ★ Optimal	Reject Very High (Score ≥ 6) + Conditional Medium	1,333 (10.3%)	57 (32.8%)	0.91%	₹32.5L	₹156.4L	Best balance: NPA reduction vs revenue

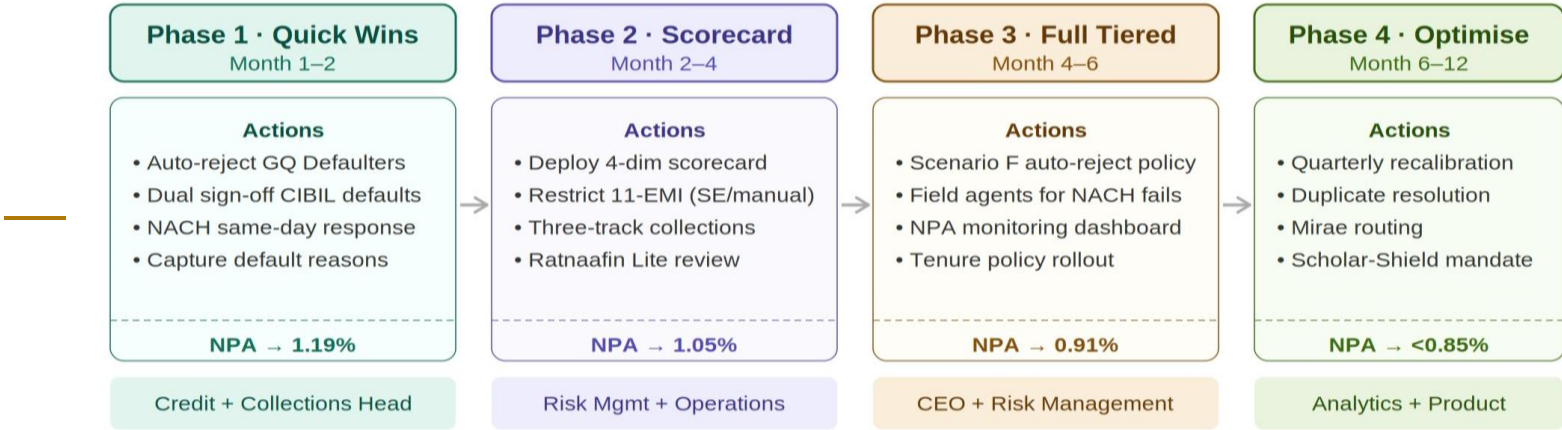
Recommendations and Impact

Recommendations — Actions Across 4 Domains



Roadmap and Impact

12-MONTH IMPLEMENTATION ROADMAP



Business Impact

Scenario analysis indicates the potential to reduce NPA exposure by approximately ₹32.5 lakh per cycle using existing operational data and infrastructure.

1.34% → 0.91%

Projected NPA Rate Under Scenario F

₹55.4 Lakh

Potentially Avoidable NPA Exposure (Manual Approvals)

57

Estimated Defaults Avoided Per Cycle

Zero

Additional Infrastructure Requirement

₹32.5 Lakh

Projected NPA Exposure Reduction / Cycle

6 Months

Estimated Implementation Timeline

Thank You